

8 The Dendritic Cytoskeleton as a Computational Device: An Hypothesis

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Summary. This chapter presents a molecular-dynamical description of the functional role of cytoskeletal elements within the dendrites of a neuron. Our working hypothesis is that the dendritic cytoskeleton, including both microtubules (MTs) and actin filaments plays an active role in computations affecting neuronal function. These cytoskeletal elements are affected by, and in turn regulate, ion-channel activity, MAPs and other cytoskeletal proteins such as kinesin. A major hypothesis we advance here is that the C-termini protruding from the surface of a MT can exist in several conformational states, which lead to collective dynamical properties of the neuronal cytoskeleton. Further, these collective states of the C-termini on MTs have a significant effect on the ionic condensation and ion-cloud propagation that have physical similarities to those recently found in actin filaments. Our objective is to provide an integrated view of these phenomena in a bottom-up scheme. We outline substantial evidence to support our model and contend that ionic wave propagation along cytoskeletal structures impact channel function, and thus the computational capabilities of the dendritic tree and neuronal function at large.

8.1 Introduction

8.1.1 Neurobiological Introduction

The neuron is the quintessential communicating cell. The strength of information processing performed by a neural circuit depends on the number of interneuronal connections or synapses. Neurons are highly polarized cells, whose level of morphological complexity increases during maturation into distinct subcellular domains. Morphologically speaking, a typical neuron presents three functional domains. These are the cell body or “soma” containing the nucleus and all major cytoplasmic organelles, a single axon, which extends away from the soma and takes cable-like properties, a variable number of dendrites, varying in shape and complexity, which emanate and ramify from the soma. The axonal terminal region, where the synapse or contact to other cells takes place, displays a wide range of morphological specializations, depending of the target area.

Turning morphology into a functional event, the soma and dendritic tree are the major domains of receptive (synaptic) inputs. Thus both the dendritic arborization and the axonal ramifications confer a high level of subcellular

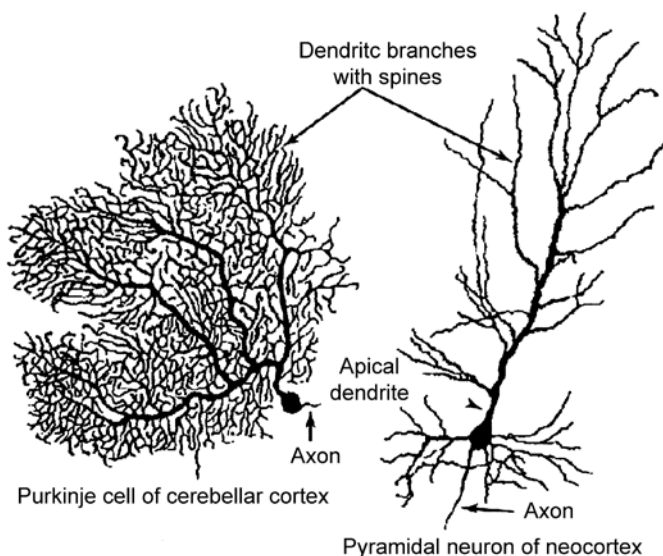


Fig. 8.1. Typical morphology of projection neurons. Reproduced from Fig. 3.1, *Fundamental Neuroscience Book*. High degree of polarization in neurons, with an extensive degree of branching, including spiny apical dendrite, shorter basal dendrites, and a single axon emerging from the basal pole of the cell. The complexity in the level of arborization is associated with the number and complexity of connections to which a neuron is associated

specificity in the localization of particular synaptic contacts on a given neuron. The extent to which a neuron may be interconnected largely depends on the three-dimensional spreading of the dendritic tree (see Fig. 8.1).

Dendrites are the principal element responsible for both synaptic integration and for the changes in synaptic strengths that take place as a function of neuronal activity [101, 46, 81]. While dendrites are the principal sites for excitatory synaptic input, little is still known about their function. The activity patterns inherent to the dendritic tree-like structures such as in integration of synaptic inputs, is likely based on the wide diversity of shapes and sizes of the dendritic arborizations [101, 49]. The size and complexity of dendritic trees increase during development [105], which has been, in turn, associated with the ability of the neural system to organize and process information, such as when animals are reared in complex sensory environments [35]. Thus, dendritic size and branching patterns (see Fig. 8.2) are important features of normal development and function. Most synapses, either excitatory or inhibitory, terminate on dendrites, so it has long been assumed that dendrites somehow integrate the numerous inputs to produce single electrical outputs. It is increasingly clear that the morphological functional properties of dendrites are central to their integrative function [79, 53]. Branching in this context is essential to the increasingly complex ability of the neuron to

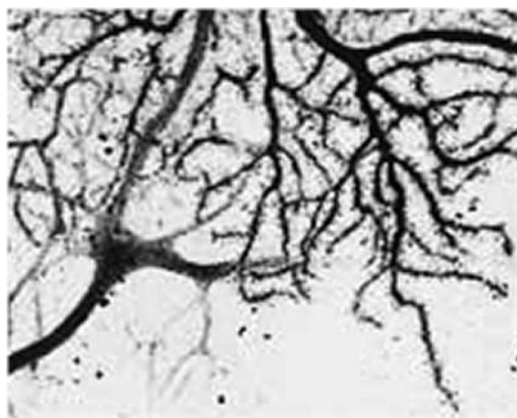


Fig. 8.2. Visualization of individual dendrites. Complex branching of the dendritic tree in Purkinje cell.

respond to developmental complexity [49]. The dendritic cytoskeleton plays a central role in the process of ramification, filopodia formation, and more specialized neuronal activity such as long-term potentiation (LTP) and long-term depression (LTD), circuit dynamics, including memory formation, and the response to pathological conditions.

Synaptic contacts take place in dendritic spines, which are small sac-like organelles, projecting from the dendritic trunk. Dendritic spines are more abundant in highly arborized cells such as pyramidal neurons and scarcer in lowly interconnected smaller-sized dendritic tree interneurons. The number of excitatory inputs can be linearly correlated with the number of spines present in the dendrite. How information is processed in a dendritic spine, is still a matter of current study [78, 141]. Dendritic spines contain ribosomes and cytoskeletal structures, including actin, and α - and β -tubulin. As in other parts of the neuronal structure, cytoskeletal components are highly relevant in the structure/function of the neuron. The axonal hillock contains large “parallel” bundles of microtubules. Axons are also located with other cytoskeletal structures, such as neurofilaments, which are more abundant in axons than in dendrites. The axon hillock is the region where the action potential is generated. The dynamics in cytoskeletal structures is central to our contention that information can be processed and “delivered” to the synaptic function by changes in the cytoskeleton structures.

It is widely accepted that chemical and structural modifications in dendritic spines underlie much of the plastic changes in the brain in response to learning and experience [142, 39]. Morphological changes in the dendritic spine are key to its plasticity and processing [78, 122]. Conversely, electrical stimulation leading to LTP, leads to the formation of new dendritic spines [27, 132]. In this regard, dynamic changes of the actin cytoskeleton have long been implicated in a number of neuronal functions, including the development and

stability of dendritic spines [28, 79, 63, 29]. Twitching of dendritic spines [29, 24] for example, which has been hypothesized to encode ultrashort memory [18], is a phenomenon that involves actin dynamics, and is blocked by cytochalasin treatment [29, 24]. The actin cytoskeleton plays a central role in synapse formation and maintenance [117]. Treating of cultured hippocampal neurons with the actin polymerization inhibitor latrunculin, for example, modifies synapse formation and stability [144]. Conversely, neuronal function also helps remodel the actin cytoskeleton. Repeated tetanic stimulation of hippocampal neurons causes remodeling of presynaptic actin, which results in the formation of potential new synapses capable of active vesicle recycling. Interestingly, even a single tetanus results in rapid translocation of actin in both presynaptic and postsynaptic compartments, such that presynaptic actin advanced towards and postsynaptic actin moved away from stimulated synapses [15]. Pruning is also associated with actin dynamics [110, 88, 138]. This process where exuberant axonal and dendritic ramification are effectively

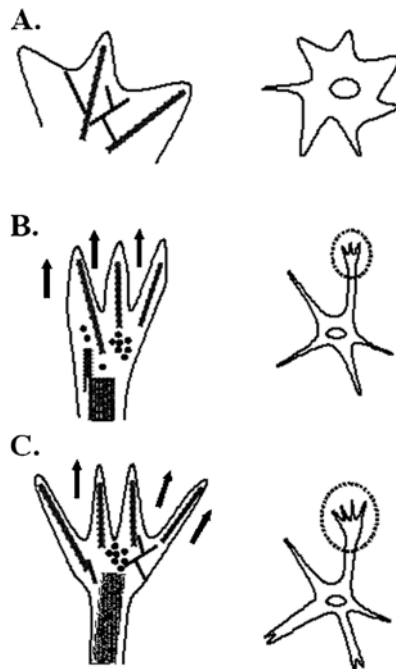


Fig. 8.3. Cytoskeletal dynamics in neurite initiation (A), neuritogenesis (B), and ogenesis (C). Reorganization of the lamellipodia with accumulation of microtubules into an ordered array results in formation of a neurite with a growth cone. The differentiation of a neurite growth cone (B) into an axon growth cone (C) requires depolymerization and reorganization of actin networks. Microtubules invade to form the characteristic central core. Actin filament dynamics is essential to all these stages of differentiation. Reproduced from [111]

eliminated is observed in developmental stages in insects where reorganization of their nervous system between larval and adult life takes place [133, 129, 64].

Thus, cytoskeletal alterations may reflect changes of neural circuits in response to learning and experience, and they must involve both destructive and constructive mechanisms (see Fig. 8.3).

8.1.2 Neuro computational Introduction

The prevailing view of long-term memory and information processing in the central nervous system continues to be heavily influenced by Hebb's conjecture [37]. The idea, which is based on the processing of information in a network of neurons, involves activity-dependent strengthening of synaptic connections between neurons. Thus, the biological steps that coordinate and strengthen synaptic activity are essential to better modeling of biological neural-based networks. McCullough and Pitts [80] showed that a sufficiently large network of simple neurons (modeled by point-like nodes – with linear threshold units) is capable of universal computation, i. e. is equivalent to a Turing machine. In this model, synaptic inputs are linearly added into a single threshold function. In the remainder of this chapter, we shall review two main routes in neural modeling, both relevant to our proposed revised model. The first deals with biologically realistic models of electrical (and chemical) signaling in the neural membrane and its implications on the way neurons respond to synaptic activity. The second approach focuses on constructing abstract models and deriving general results regarding their computational capabilities using analytic and simulative methods.

Several variants of the point-like neuron model exist (see [82, 60]). The simplest is the “integrate and fire” (IF) in which charges are accumulated across the membrane (modeled by a capacitor) via synaptic inputs (positive or negative) until a certain threshold is reached, causing an instantaneous discharge and a pulse generating along the axon. The next variant, known as the “leaky IF” (LIF), modeled as an “RC” circuit, includes a resistance in parallel to the membrane capacitor. In this case there is a certain discharge at all times, such that the membrane potential undergoes an exponential decay over time. Again, the neuron generates an impulse upon the event of (voltage) threshold crossing. While the variants described above are operating in an “all-or-none” fashion, the “rate neuron” has a continuous activation function that is monotonically increasing as a function of the membrane's electrical activity, reaching a saturating constant value for high-activity regimes. This squashed function represents the average firing rate of a group of connected neurons.

This highly restrictive picture effectively models as the whole dendritic arbor as a single-point compartment. However, an application of the linear cable theory to dendrites [99, 41] gave rise to several interesting insights regarding the distribution of the membrane potential, which clearly challenge

this naive model. According to the one-dimensional passive cable equation analysis, dendrites:

1. Impose a large conductance load on the soma,
2. Low-pass filter postsynaptic potentials (PSPs),
3. Change the response at the soma to various combinations of synaptic activities [100].

However, while most of these models assume passive membrane properties, it is clear that the dendritic electrical properties change dynamically by modulation of a variety of voltage-gated ion channels [44, 20], and by dynamic changes in its cytoskeletal components [30, 73, 5, 32]. Dendritic channels provide this compartment with the ability to perform nonlinear operations on PSPs, at least locally. This means that synapses may interact in a nontrivial way, a fact that gives rise to a completely new set of possibilities regarding the operation and the function of dendrites [81, 119, 60]. Several types of experimental evidence support this view, hence weakening the relation between stable synaptic weights and long-term memory, as well as learning, namely:

- Neurons demonstrate significant morphological plasticity both by growing dendritic branches (even in mature cells) and by rapidly developing new dendritic spines [39]. This type of evidence supports alternative mechanisms for memory acquisition and handling that extend beyond the classical Hebbian conjecture.
- Experiments have revealed significant fluctuations of synaptic efficacy over short periods of time in response to a recent burst of activation, e. g. [135, 38]. This type of evidence questions the conventional assumption of stable synaptic strength whose fine resolution is obtained via repeated presentation of a set of training examples. This assumption is also one of the building blocks of most connectionist models [7].
- The dendritic membrane expresses a variety of active ion channels (NMDA receptors, Na^+ and Ca^{2+} and other voltage-dependent channels [115, 49] that can significantly affect the electrical properties, and hence integrative behavior of this compartment. Dendrites thus exhibit a capability to amplify their synaptic input. At the same time, nonlinear responses have been localized to specific dendritic arbors and even to single branches within a dendritic domain, clearly pointing to compartmentalization mechanisms capable of confining certain inputs to a specific dendritic region.

The above findings have triggered a revised view of information processing in dendrites. Rather than having a passive membrane, nonlinear models were proposed to incorporate the above evidence [60]. These adaptive models demonstrate that a novel and wider repertoire of capabilities than previously expected from large networks of simple integrating units (even with passive dendrites) are required, which are not supported by conventional models. Among these new capabilities one can count, multiplication, fast correlation/coincidence, which is beyond the capabilities of interneuron

transmission due to long temporal delays, and reduced sensitivity of somatic excitatory-PSP to the spatial distance from the dendrite [60]. In this context, dynamic cytoskeletal remodeling and functional interactions between cytoskeletal structures and synaptic ion-channel activity may be at the center of this revised model of neuronal adaptability [56, 32].

8.1.3 Dendritic Channel Function

Although action potentials can initiate and propagate in dendrites of various neurons, very little information was available until recently concerning the ion channels that sustain them. Ca^{2+} -dependent action potentials have been observed in hippocampal neurons and dendrites [116, 6]. Firing of action potentials elicits characteristic M-shaped Ca^{2+} profiles across the neuron. These spatially constrained Ca^{2+} signals are essential to distinct neuronal function. The Ca^{2+} rise is small in the soma, highest in the proximal apical and basal dendrites, and small again in distal apical dendrites [43, 83, 102–104]. Interestingly, addition of the K^{+} -channel blocker TEA renders the Ca^{2+} distribution more uniform, with large increases that can be observed at the very tips of the dendrites [43]. These findings originally suggested the presence of voltage-gated Ca^{2+} channels in the dendritic tree. The spatial distribution of Ca^{2+} channels may reflect the propagation into the dendrites of trains of action potentials. Imaging with Na^{+} -sensitive dyes has also yielded similar M patterns, suggesting that Ca^{2+} influx into the dendrites is driven by Na^{+} -dependent action potentials [43, 44]. Both Na^{+} -channel inactivation and K^{+} -channel activation during action potentials would only allow a first few spikes, not necessarily sufficient to actively propagate the entire extent of the dendrites [43].

8.1.4 Actin–Microtubule Cytoskeletal Connections

Classically, F-actin and microtubular cytoskeletal networks are thought to fulfill separate, independent cellular roles. Highly dynamic actin networks are known for their role in cell spreading and contraction. The more stable microtubular cytoskeleton is best known for its importance in cell division and organelle trafficking. However, recent studies provide a more unified role, ascribing important roles for the actin cytoskeleton in cell division and trafficking and conversely, important roles for microtubules in the generation and plasticity of cellular morphology. Coordination between the actin- and microtubule-based cytoskeletons has also been observed during the process of neurite initiation [67, 21], which shares similarities with cellular migration and morphogenesis. A direct physical association between both cytoskeletons has been suggested, because microtubules often preferentially grow along actin bundles and transiently target actin-rich adhesion complexes. In neurons, certain plakins and neuron-specific microtubule-associated proteins (MAPs), including MAP1B and MAP2, play a role linking microtubules and F-actin, and

helping the transition from an undifferentiated state to neurite-bearing morphology [21]. Studies on contact-induced neurite formation in chick cortical neurons, however, suggested that microtubule polymerization is not required for the generation of neurites [123]. In this study, neurite-like protrusions were formed by the occasional contact of filopodia between two neurons on a coverslip. Upon contact, these filopodia grew in diameter and became filled with microtubules. Inhibition of microtubule dynamics by the combined application of the microtubule-stabilizing drug taxol and the destabilizing drug nocodazole did not prevent protrusion formation, which became filled with microtubules. This suggests that microtubules do not grow, but instead are transported into such locations. Depolymerization of most pre-existing microtubules with nocodazole induced cytosol flow into filopodia; which resulted in short and thick protrusions that did not resemble neurites in many respects. Thus, polymerization of new microtubules may not be a requirement for microtubule reorganization during contact-mediated neurite formation. In spontaneous neurite initiation, in contrast, microtubule polymerization does appear to be essential. Both taxol [67] and nocodazole [21], treatment, which inhibit microtubule polymerization and depolymerization, respectively, affected neurite formation without affecting pre-existing microtubules [143, 106, 68]. However, in the absence of normal microtubule polymerization, processes fail to form growth cones and fail to generate a thin neurite shaft. This suggests that key events in spontaneous neurite initiation, such as the separation of the lamellipodium from the cell soma to become a growth cone, are dependent on both the presence and the dynamic properties of microtubules. Paradoxically, experimental evidence suggests that actin filaments can both inhibit and promote neurite initiation and outgrowth [31]. Robust neurite-like induction was observed in neuroblastoma cells incubated with a variety of actin-disrupting drugs such as cytochalasin D, latrunculin A, and jasplakinolide, in combination with microtubule stabilizing concentrations of taxol [22]. These observations and others [26, 57] indicate that the actin cytoskeleton may be dispensable in process initiation, and in some cases the actin network is actually inhibitory. One observation that has emerged in studies of growth cones [114] is that actin filaments and microtubules are in intimate contact at the cell periphery. This suggests that membrane-associated complexes may directly tether both cytoskeletal systems. Growth cones are organized into distinct domains: a microtubule-rich central domain (C-domain) and an actin-rich peripheral domain (P-domain) [66, 69]. Although microtubules are only rarely observed within the P domain, over time many microtubules alternately advance into and retreat from the P-domain [127]. This can be observed in the lamellipodia of morphologically undifferentiated neurons prior to neurite initiation [22]. Certain perturbations, like exposure to actin assembly inhibitors, result in microtubular invasion into the P-domain [31]. Thus, it is possible that actin filaments restrain the microtubule cytoskeleton, as well as organelles and cytoplasm, from advancing toward the leading edge, such as in the actin-based retrograde flow [136, 109, 114].

Thus, dynamic interactions between both actin and microtubular cytoskeletons may be central to neuronal cell morphology and function. Only recently have clear functional interactions between those cytoskeletal structures became apparent. Association of microtubules with actin cables in filopodia appears to guide microtubule growth along the most efficient path antiparallel to retrograde flow [114]. Thus, specific coupling between microtubules and actin bundles possibly promotes microtubule advance. Association of microtubules with actin cables in lamellipodia of neuroblastoma cells expressing recombinant MAP2c, further suggests direct actin–microtubule interactions would play a key role in neurite initiation [22]. Actin–microtubule interactions might also be important for growth cone turning underlying axonal path-finding mechanisms. Both actin filament bundles [11] and dynamic microtubules [12] are required for growth-cone turning to avoid inhibitory cues. Increased microtubule invasion into neuronal lamellipodia might be essential for axon branching [23] and neurite initiation [22]. This is further evidenced by the presence in neurons of proteins capable of interacting with both microtubules and F-actin, and proteins that can mediate signaling between both cytoskeletons, which are thus likely to control microtubular invasion. The microtubule-associated proteins MAP1B and MAP2, for example, are known to interact with actin *in vitro* [96, 131, 113, 120, 17]. The C-terminal domain of the L2 subunit of MAP1B was shown to bind to F-actin and interact with stress fibers in transfected cells [131, 87]. Furthermore, MAP2 binds to F-actin efficiently, and induces actin bundle formation [113, 36]. MAP2 has also been found to localize in regions devoid of microtubules in neurons [84, 9]. Under certain conditions such as PKA activation, for example, MAP2 is found to be associated with the actin cytoskeleton [92]. This suggests that actin-related functions of MAP2 are regulated by phosphorylation. The MAP2-related MAP tau has been proposed to interact and bundle F-actin as well [36]. It is likely that by crosslinking, MAP2 and/or MAP1B associates with both cytoskeletons, and could thus be involved in guidance of microtubules along actin filament bundles.

8.2 Collective Modes of Behavior of C-Termini in Microtubules

In the following section we focus our attention on a specific dynamic mode of behavior within the MT cytoskeleton of a dendrite. MTs are long hollow cylinders made of $\alpha\beta$ -tubulin dimers [25]. These structures have outer diameters measuring 25 nm and inner diameters of 15 nm. During cell division, tubulin can account for as much as 10% of the cell's protein. MTs form by polymerization of tubulin dimers in a GTP-dependent process. In some cases, strands or protofilaments are formed first. MTs typically consist of 13 such protofilaments. Each μm of MT length consists of 1650 heterodimers. Recently, it has become apparent that neurons utilize MTs in cognitive processing. Both

kinesin and MAP2 have been implicated in learning and memory [139, 54, 140]. Dendritic MTs are implicated in particular, and it is highly probable that precisely coordinated transport of critical proteins and mRNAs to the postsynaptic density via kinesin along MT tracks in dendrites is necessary for learning, as well as for long-term potentiation (LTP) [55, 126]. As shown schematically in Fig. 8.4, presynaptic axon terminals make synaptic contact onto postsynaptic spines. Within the shaft of the dendrite, MTs act as tracks for kinesin motor protein propagation [2]; thus enabling them to transport proteins such as receptors. MTs also transport mRNA that translates proteins critical to synaptic activity.

In the following section we focus on the C-termini of neighboring tubulins, whose biophysical properties have a significant influence on the transport of material to activated synapses, thus affecting the capability of signal transduction and processing at the cytoskeleton on the one hand, and synapses undergoing LTP on the other hand. Using molecular dynamics modeling we calculated conformation states of the C-termini located on the outside of the MT, strongly interacting with other proteins, such as MAP2 and kinesin [108]. The processivity of kinesin appears to critically involve an interaction with the C-termini of tubulin, although MAP2 may temporarily uncouple to avoid interfering with the movement of kinesin along the MT [130]. On the other hand, removal of the C-terminus of tubulin profoundly affects kinesin processivity. Cleavage by subtilisin, for example, decreases the rate of kinesin transport by four-fold [137].

To elucidate the biophysical properties of C-termini and gain insight into the role it plays in the functioning of dendrites, we developed a quantitative computational model based on the currently available biophysical and

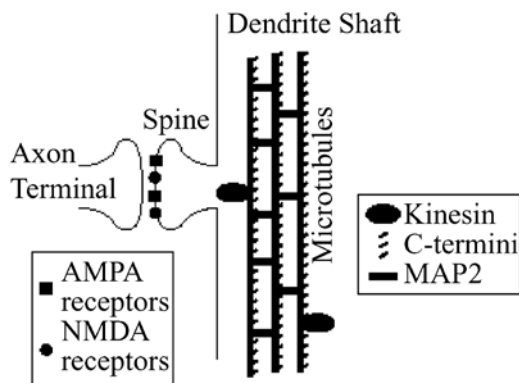


Fig. 8.4. Diagram of C-termini on the tubulin dimers of microtubules in relation to incoming synapses. Axon terminals typically input to spines, many of which contain AMPA and NMDA glutamate receptors. Kinesin motors transport cargo along microtubules. C-termini states affect the transport of kinesin along microtubules and this affects transport of materials to active synapses

biochemical data regarding the key macromolecular structures involved, including tubulin, their C-termini, and associated MAP2. In the following we describe the potential configurations of C-termini based on:

- The interactions between C-termini and the tubulin dimer,
- The interactions between nearest-neighbor C-termini, and
- The interactions between C-termini and MAP2 that lead to a model of wave propagation through counterions bounded to MAP2.

Our bead-spring model of the C-termini extends the static results, providing a finer picture of the dynamic behavior, the conformation states and their distribution.

8.2.1 Potential Configurations of Microtubular C-Termini

Based on the assumption that C-termini dynamics plays an important role in the functioning of a dendrite, we developed a simplified model of a C-terminus and its interactions with neighboring structures. The proposed model of the C-termini microtubular network is schematically illustrated in Fig. 8.5, where the tubulin dimer is considered to be the basic unit. Each dimer is decorated with two C-termini that may either extend outwardly from the surface of the protofilament or bind to it in one of few possible configurations (see next section).

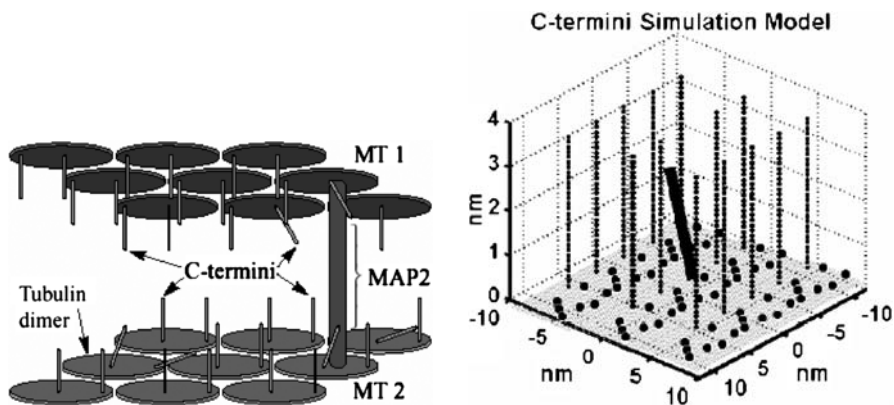


Fig. 8.5. *Left:* A schematic diagram of portions of two microtubules connected by a MAP2. Each portion shows the outer face of a tubulin dimer covered by two C-termini. Each C-terminus can be in one of two metastable states, perpendicular or parallel to the dimer surface. *Right:* A portion of the microtubule composed of a few tubulin dimers with their C-termini used as a model for the static simulation (*right*). Also shown are a few of the positively charged regions (circles) on the surface of a tubulin dimer. The test C-terminus is slanted (bold)

The most dynamic structural elements of the system (i.e. its elastic and electric degrees of freedom) can be best understood in terms of the conformational states of the C-termini. Each state of the unbound C-terminus is assumed to evolve such as to minimize the overall interaction energy of the system. According to our model, the C-termini, which are negatively charged, interact electrostatically with (a) the surface of the dimer, (b) the neighboring C-termini and (c) an adjacent MAP. Note that while the surface of the dimer is highly negatively charged on the average, it exhibits regions of positive charge that attract the C-termini causing them to bend and bind in what we generally denote as the “downward” state. The energy difference between the two major metastable states is relatively small, on the order of a few $k_B T$ at room temperature.

A MAP of the interaction energy of a C-terminus and its major neighbors reveals its static properties (the dynamic properties will be revisited later). For this purpose we studied numerically the interaction energy surface as experienced by a C-terminus. The details of the model and the terms involved are given in [98]. Fig. 8.6 shows a surface plot of the energy. It transpires that the “up” state has the lowest energy (corresponding to the C-terminus being perpendicular to the tubulin’s surface). However, the cone angle created by the constraint $E - E_0 < 50$ meV (where $E_0 = E$, $\theta = 90^\circ$) is about 40° . This means that the C-termini can move freely within this cone due to thermal fluctuations (with $k_B T$ approximately equal to 25 meV at physiological temperature). The important result is the existence of local minima associated

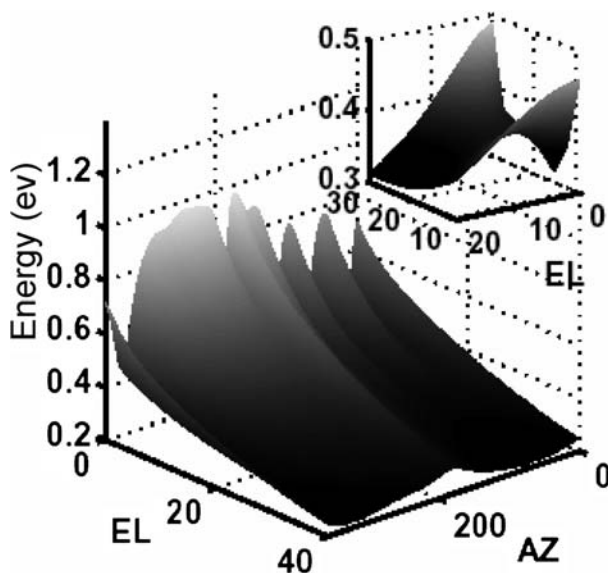


Fig. 8.6. Energy surface of the interaction between the test C-terminus and the environment. The inset shows a saddle point on the surface

with the “downward” state, which is only 100 meV higher than the straight-up state. The saddle point found is 160 meV above the “up-state”. What we deduce here is that, at least under the assumptions of the model, there are two major metastable states with a moderate energy barrier separating them. The inset in Fig. 8.6 shows a close-up view around the saddle point. In the next section we shall see that a more detailed model reveals the availability of multiple configurations associated with the “downward” state.

8.2.2 Dynamic Model of the C-Termini

To enable exhaustive simulations, we modeled the C-terminus as a sequence of beads with flexible connections, instead of the rigid rod, in what is known as a bead-spring model. The model takes into account the electric field exerted by the dimer, the external field generated by the environment, and various short-range interactions within the simulated C-terminus, including interactions between the beads (i. e. Lennard-Jones potential, angular forces, etc.). Our simulation indicates the ability of an ionic wave to trigger a corresponding wave of C-termini state changes from their upright to downward orientations as predicted by the simplified model discussed above.

Four views of C-termini states produced by this model are shown in Fig. 8.7, each has different conformation with examples of the “up” and “down” states evident. The position distribution showed in Fig. 8.8 describes the minimal energy positions of beads representing the constituent residues of a C-terminus. Calculations of the energy-minimized positions of the individual beads representing the amino acids of the C-termini in two equivalent

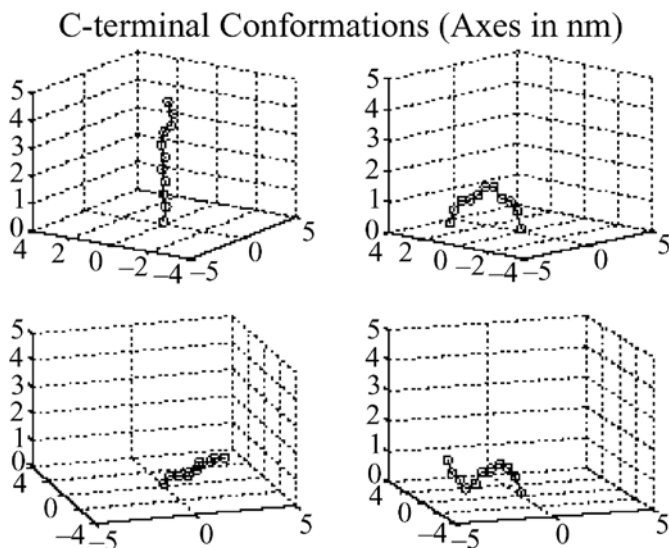


Fig. 8.7. Various C-terminal conformations observed with the dynamic model

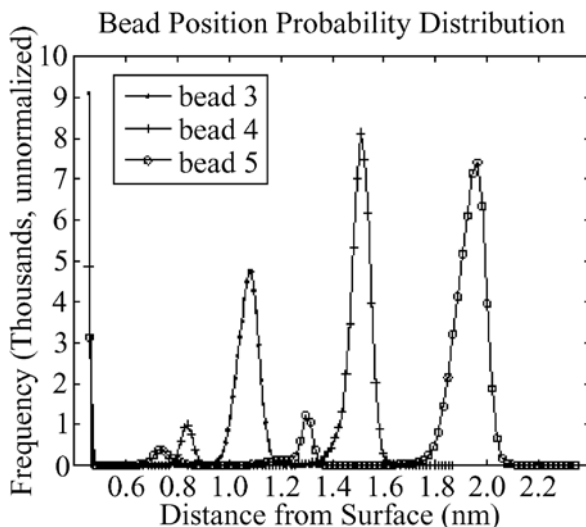


Fig. 8.8. Probability distribution describing minimal energy positions of beads representing a C-terminus ranging from 3 to 5 obtained at the end of each run after relaxation. The energy-minimized value can come from each of the beads at positions from 3 to 11. The first bead is permanently connected to the surface and hence has no freedom to move

forms reveals that the probability of the down position that includes all cases of full or partial attachment is 15%. This means that the system has indeed two major states with a strong bias towards the stretched-up state. In other words, the system tends to be mostly in the “up-state” unless driven towards one of the surface binding sites where the probability of attraction increases.

8.2.3 Ionic Wave Propagation along MAP2

Years ago, Manning [77] postulated an elegant theory, which stated that polyelectrolytes in solution may have condensed ions in their surroundings. According to Manning’s hypothesis, counterions “condense” about the polymer’s length, provided that a sufficiently high linear charge density is present on its surface. Thus, a linear polymer may be surrounded by counterions from the saline solution such that counterions are more closely surrounding the polymer’s surface, and salt co-ions in the solution are repelled such that a depletion region is also created [77, 93]. Although this theory was originally postulated for such polyelectrolytes as DNA, the same applies to highly charged one-dimensional polymers such as MAPs.

In an effort to investigate the ability of MAP2 to function as a “waveguide” that transfers the conformational change in a C-terminus state to an adjacent MT, we studied a simplified model of the interaction between MAP2 and its ionic environment via counterions. While the 3D structure of MAP2 is

unknown since this protein has not been crystallized, it has been investigated for its binding with MTs. Although it is still uncertain whether a single MAP2 makes direct physical contact with a MT, it is apparent that it at least reaches its immediate vicinity. Our calculations of the charge distribution on the surface of the projecting domain of MAP2 resulted in a total negative charge of 150 e and a very close to linear charge distribution. To simplify the model we assumed that the counterion-attracting sites are equidistant and are

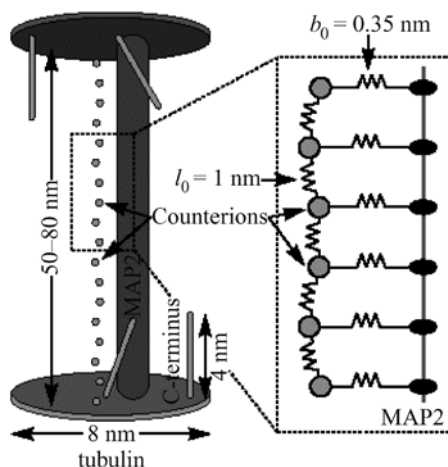


Fig. 8.9. Schematic model of a MAP2 interacting with positive counterions (*circles*). Also seen are two dimers at either end of the MAP2 and a slanted C-terminus. The zoomed region depicts the springs model used to describe the interactions between the MAP2 and the counterions

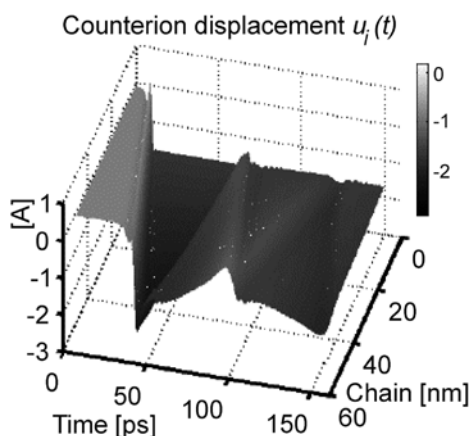


Fig. 8.10. Results of the MAP2 model simulation describing the counterions displacement (the chain-axis denotes the discrete indices of the counterions). The color bar is given in [Å]

arranged on a straight chain along the MAP2, schematically shown in Fig. 8.9. Hence we envisage that the counterions move only in a plane perpendicular to a cylinder representing the MAP2. We show that a perturbation applied to the counterions at one end of the MAP2 will drive them out of equilibrium and initiate a wave that travels along the MAP2. Details of the model are given elsewhere [98].

Fig. 8.10 depicts the main result of this analysis for a localized perturbation applied for a few picoseconds to the counterions near the binding site of the MAP2. The wave propagation along the chain of $N = 50$ counterions is represented by the counterion displacement parallel to the MAP2, u_i , where i denotes the i -th counterion. The profile of the displacement reveals that the perturbation along MAP2 propagates almost as a “kink”. Measurements of the phase velocity give values in the order of $v_{\text{ph}} \approx 2 \text{ nm ps}^{-1}$.

8.3 Ion Waves along Actin Filaments

8.3.1 Ionic Condensation along the Actin Filament

Motivated by several intriguing experiments [10, 70] indicating the possibility of ionic wave generation along actin filaments, we have also developed a physical model that provides a framework for the analysis of these waves. Actin filaments are highly charged one-dimensional polymers with an uneven distribution of charges along the polymer’s length. As a consequence, spatially dependent electric fields could be present and arranged in peaks and troughs as originally postulated by Oosawa [90]. This might imply large changes in the density of small ions around the polymer with a large dielectric discontinuity in the ionic distribution [51, 42, 3]. This uneven ionic distribution along a short stretch of the polymer (likely the average pitch 35–40 nm) may be considered a linear unit of an electrical circuit to be modeled with nonlinear electrical components. First, there is a nonlinear capacitor, which is associated with the spatial charge distribution between the ions located in the outer and inner regions of the polymer. Secondly, any charge movement tending to dissipate this local gradient (reversibly) generates a secondary electromotive force due to the charge movement, namely a local current. This is the main contribution to the inductance of the single stretch of polymer. F-actin, being a highly charged polyelectrolyte contains a fraction of its surrounding counterions in the form of a condensed cloud about its surface. Such a cloud may be highly insensitive to large changes in the ionic strength of the surrounding saline solution [89, 76, 145]. The Bjerrum length, λ_B , describes the distance beyond which thermal fluctuations are stronger than the electrostatic attraction or repulsion between charges in solution whose dielectric constant is $[\varepsilon]$. It is defined by

$$\frac{e^2}{4\pi\varepsilon\varepsilon_0\lambda_B} = k_B T, \quad (8.1)$$

for a given temperature T in Kelvin. Here e is the electronic charge, ε_0 the permittivity of the vacuum and k_B is Boltzmann's constant. For a temperature of 293 K it is readily found that $\lambda_B = 7.13 \times 10^{-10}$ m. Counterion condensation occurs when the mean distance between charges, b , is such that $\lambda_B/b = S > 1$. Each actin monomer carries an excess of 14 negative charges in vacuum, and accounting for events such as protonation of histidines, and assuming there to be 3 histidines per actin monomer, there exist 11 fundamental charges per actin subunit [128]. Assuming an average of 370 monomers per μm we find that there is approximately $4e/\text{nm}$ in agreement with an earlier statement. Thus we expect a linear charge spacing of $b = 2.5 \times 10^{-10}$ m, such that $S = 2.85$. As the effective charge, q_{eff} , or renormalized rod charge is the bare value divided by S we find $q_{\text{eff}} = 3.93e/\text{monomer}$.

Assuming for simplicity a linear charge distribution about the actin filament, the linear charge density, ξ , is much greater than $1/z$ where z is the valence of the counterions in solution. The parameter, ξ , is given by

$$\xi = \frac{e^2}{4\pi\varepsilon_0\varepsilon_r\lambda_B k_B T b}, \quad (8.2)$$

where e is the electronic charge, ε_r the dielectric constant of water, and b the average axial spacing of charges on the polyelectrolyte [145]. The parameter, ξ , was calculated to be 110 for actin filaments (with $z = 1$ for H^+ and K^+ ions) and so $\xi \gg 1/z$ [70]. Thus approximately 99% of the counterion population is predominantly constrained within a radius of 8 nm [97] round the polymer's radial axis [145]. Significant ionic movements within this "tightly bound" ionic cloud are therefore allowed along the length of the actin (Fig. 8.11) provided that it is shielded from the bulk solution [89, 93].

Bearing in mind the sheath of counterions around the actin filament we see that effectively actin polymers may act as biological "electrical wires" [70], which can be modeled as nonlinear inhomogeneous transmission lines. Such electrical devices are able to propagate nonlinear dispersive solitary waves [61, 91] in the form of solitons [86, 71, 72]. It has been proposed that such solitons, localized traveling waves, exist in many biological systems and, in fact, have earlier been postulated in linear biopolymers such as MTs [112] and DNA [4]. Interestingly, both solitary waves [70] and liquid crystal formation [16, 33] have been observed in actin filaments possibly providing a potentially fruitful environment for innovative biotechnological applications.

8.3.2 Electrical Modeling of Actin

We developed a model based on the transmission-line analogy with inductive, resistive and capacitive components. The physical significance of each of the components, for each section of the electrical network, as well as additional details is described in [134]. As the counterion condensation cloud separates the filament core from the rest of the ions in the bulk solution, we expect

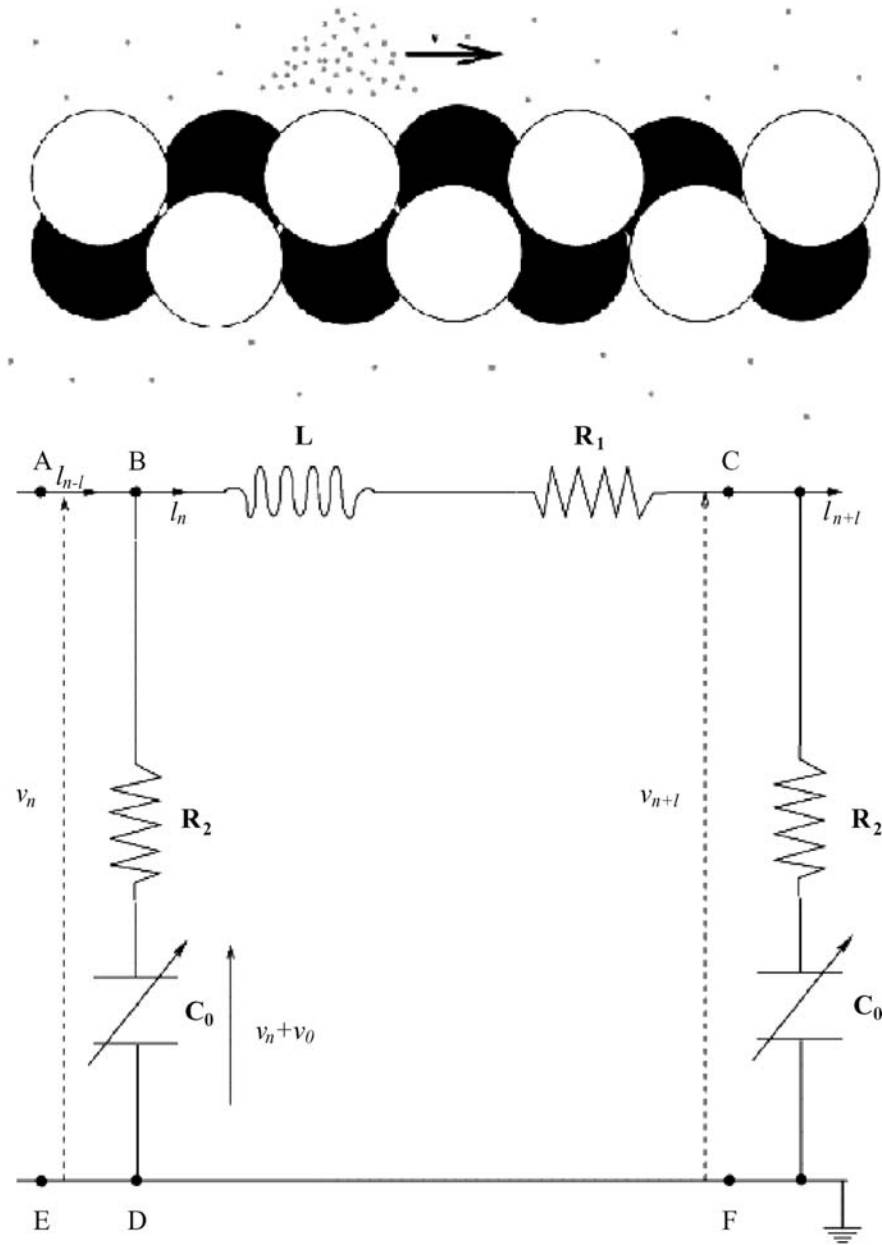


Fig. 8.11. Nonlinear dispersive solitary waves traveling along an actin filament. Positive counterionic charges in the bulk form a cylinder around the actin filament. The distance between this cylinder and the actin filament is one Bjerrum length. There are also some counterions that condense to the actin filament. An effective circuit diagram for the n -th monomer between the dotted lines. I_n is the current through the inductance L and resistance R_1

this cloud to act as a dielectric medium between the two. This cloud provides both resistive and capacitive components for the behavior of the monomers, which make up the actin filament. Ion flow is expected at a radial distance from the center of the filament, which is approximately equal to the Bjerrum length. The inductive component is due to the actin's double-stranded helical structure that induces the ionic flow in a solenoidal manner.

For actin in solution, a key feature is that the positively charged end assembles more quickly than the negatively charged end [121]. This results in an asymmetry in the charges at the ends of the filaments and F-actin's electric polarization. Actin monomers arrange themselves head to head to form actin dimers resulting in a probable alternating distribution of electric dipole moments along the length of the filament [58, 59]. We assume, therefore, that there is a helical distribution of ions winding around the filament at approximately a radial distance equal to one Bjerrum length. This may be thought of as a solenoid in which a fluctuating current is flowing as a result of voltage differences generated at each end. The number of turns, $N = a/r_h$, is approximated by simply working out how many ions could be lined up along the length of a monomer. We would then be approximating the helical turns as circular rings lined up along the axis of the F-actin. This is surely an overestimate of the number of helical turns per monomer, but it is certainly within reason for our purposes as an initial approximation. We also take the hydration shell of the ions into account in our calculation. This shell results when an ion is inserted into a water configuration, changing the structure of the hydrogen-bond network. A water molecule will reorient such that its polarized charge concentration faces the opposite charge of the ion. As the water molecules orient themselves towards the ion, they break the hydrogen bonds to their nearest neighbors. The hydration shell is then the group of water molecules oriented around an ion. A rough estimate for the size of a typical ion in our physiological solution, with a hydration shell, is $r_h = 3.6 \times 10^{-10}$ m. This is the approximate radius of the first hydration shell of sodium ions. For a filament containing n monomers, we would get an effective resistance, inductance, and capacitance, respectively, such that:

$$R_{\text{eff}} = \left(\sum_{i=1}^n \frac{1}{R_{2,i}} \right)^{-1} + \sum_{i=1}^n R_{1,i}, \quad (8.3)$$

$$L_{\text{eff}} = \sum_{i=1}^n L_i, \quad (8.4)$$

and

$$C_{\text{eff}} = \sum_{i=1}^n C_{0,i}, \quad (8.5)$$

where $R_{1,i} = 6.11 \times 10^6 \Omega$, and $R_{2,i} = 0.9 \times 10^6 \Omega$, such that $R_{1,i} = 7R_{2,i}$. The reader should note that we have used $R_{1,i} = R_1, R_{2,i} = R_2, L_i = L$, and $C_{0,i} = C_0$. For a 1- μm length of actin filament we find therefore

$R_{\text{eff}} = 1.2 \times 10^9 \Omega$, $L_{\text{eff}} = 340 \times 10^{-12} \text{ H}$, $C_{\text{eff}} = 0.02 \times 10^{-12} \text{ F}$. The electrical model of the actin filament is basically an application of Kirchhoff's laws to that section of the effective electrical circuit for one monomer, M , which involves coupling to neighboring monomers. We then take the continuum limit, assuming a large number of monomers along an actin filament. The application of Kirchhoff's laws and using the continuum approximation the following equation is derived [134], which describes the spatiotemporal behavior of the potential along the actin filament:

$$LC_0 \frac{\partial^2 V}{\partial t^2} = a^2 (\partial_{xx} V) + R_2 C_0 \frac{\partial}{\partial t} (a^2 (\partial_{xx} V)) - R_1 C_0 \frac{\partial V}{\partial t} + R_1 C_0 2bV \frac{\partial V}{\partial t}. \quad (8.6)$$

As a result of applying an input voltage pulse with an amplitude of approximately 200 mV and a duration of 800 μs to an actin filament, electrical signals were measured at the opposite end of the actin filament [62]. This experiment indicates that actin filaments can support ionic waves in the form of axial nonlinear currents. The current measured in the process lasted approximately 500 μs , and reached a peak value of approximately 13 nA. In a related earlier experiment [70], the wave patterns observed in electrically stimulated single actin filaments were remarkably similar to recorded solitary waveforms from experimental studies on electrically stimulated nonlinear transmission lines [61, 71, 86]. Considering the actin filament's highly nonlinear complex physical structure [97] and thermal fluctuations of the counterionic cloud from the average distribution [89, 90], the observation of soliton-like ionic waves is consistent with the idea of actin filaments functioning as biological transmission lines.

Indeed the soliton conformation theory for proteins, involves charge displacement for "resonant" states within a particular structure as predicted by Davydov (see [19]). However, in the case of actin in solution, the soliton behavior envisaged in our model is constrained to the ion clouds and represents the "convective movement of ions" along a one-dimensional polymer.

Analysis of the nonlinear equations using two different approaches resulted in two possibilities. The first ansatz resulted in a zero dissipation state (due to the absence of resistive terms) corresponding to an absolute maximum velocity of propagation. This, however, implies a purely electromagnetic disturbance propagating resonantly from node to node in the circuit. A more physically realistic case involved mapping our equation onto the Fisher–Kolmogoroff equation with the attendant elliptic and topological solitonic solutions. The elliptic waves were stationary in time and may lead to the establishment of spatial periodic patterns of ionic concentration.

8.3.3 Implications of Actin Filament's Electrical Activity

Successful application of electrical circuit components indicates the existence of a traveling kink wave, which describes a moving transition region between high and low ionic concentrations due to the corresponding intermonomeric

voltage gradient. The velocity of propagation was estimated to range between 1 and 100 ms^{-1} depending on the characteristic properties of the electrical circuit model. Interestingly, these values overlap with action potential velocities in excitable tissues [40]. Considering the abundance of actin filaments in axons and dendritic tress, our findings may have important consequences for our understanding of the signaling and ionic transport at the intracellular level. Extensive new information (see [45]) indicates that actin filaments are both directly [13] and indirectly linked to ion channels in both excitable and nonexcitable tissues, providing a potentially relevant electrical coupling between these current generators (i.e. channels), and intracellular transmission lines (i.e. actin filaments). Furthermore, actin filaments are crucially involved in cell motility and, in this context, they are known to be able to rearrange their spatial configuration. It is tantalizing to speculate that ionic waves surrounding these filaments may participate or even trigger the rearrangement of intracellular actin networks. In nerve cells actin filaments are mainly located in the synaptic bouton region. Again, it would make sense for electrical signals supported by actin filaments to help trigger neurotransmitter release through voltage-modulated membrane deformation leading to exocytosis [118]. Actin is also prominent in postsynaptic dendritic spines, and its dynamics within dendritic spines has been implicated in the postsynaptic response to synaptic transmission. Kaech et al. [52] have shown that general anesthetics inhibit this actin-mediated response. Incidentally, it is noteworthy that Claude Bernard showed also that the anesthetic gas chloroform inhibited cytoplasmic movement in slime mold. Actin structural dynamics plays a significant role in synaptic plasticity and neuronal function. Among functional roles of actin in neurons, we mention in passing glutamate receptor channels, which are implicated in long-term potentiation. It is therefore reasonable to expect ionic wave propagation along actin filaments to lead to a broad range of physiological effects.

8.4 Dendritic Cytoskeleton Computation – Vision of Integration

In previous sections we discussed novel properties of the main components of the neuronal cytoskeleton, in particular ionic wave propagation along actin filaments, and interacting MAPs and C-termini in microtubular tubulins. This is the core of our current hypothesis. The cytoskeletal biopolymers, including actin filaments and microtubules, constitute the backbone, through possible wave propagation via those structures, which in turn interact with membrane components, including ion channels thus rendering novel modulatory effects to synaptic connections. The basic experimental finding that actin filaments support wave propagation [70] suggests the possibility of nonlinear, soliton-type propagation along “one-dimensional” biopolymers [93], provided that ionic condensation is supported [76, 89, 77]. The molecular dynamics mod-

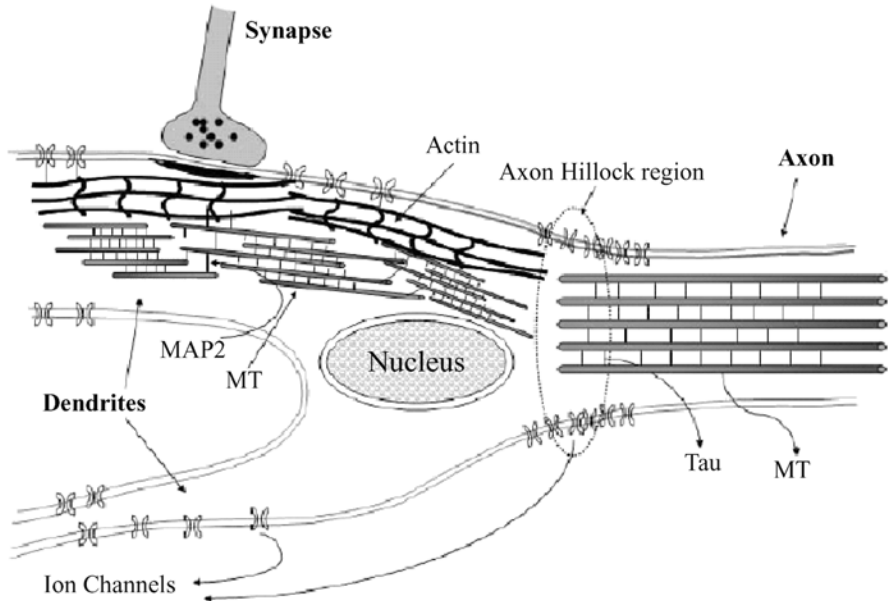


Fig. 8.12. Basic scheme of a neuron showing dendrites, the soma and an axon. Depicted within the dendrites are microtubule networks (MTs interconnected by MAP2), actin filaments and a synapse. Actin bundles and microtubules are interconnected. Actin filaments are also connected to ion channels

eling of microtubules also raises the possibility that these polymeric structures are capable of transmitting electrostatic perturbations both collectively among neighboring carboxy-termini and from these, to adjacent MTs via linking proteins such as MAP2 [95, 1]. Before we begin to describe our hypothesis, let us illustrate the basic relations between the main components of the model. Fig. 8.12 shows a general scheme of the neuron, emphasizing the main components to be discussed later. Some of the basic morphological features include the Tau-MAP interconnected bundles of microtubules (MTs) present in the axon, and the axon hillock, a region rich in ion channels. The dendritic shaft contains microtubule networks (MTN), MAP2-interconnected MTs, and actin filaments and synapses. Microtubules and actin filaments are interconnected, and actin filaments are connected to ion channels.

Fig. 8.13 focuses on the dendritic shaft. The MTs are decorated by C-termini and interconnected by MAP2 (thick line). Connections between MTs and actin filaments are shown as well. Two types of synaptic bindings are depicted. On the upper left side, actin bundles bind to the postsynaptic density (PSD) of a spineless synapse. On the lower right hand, a spiny synapse is shown where the actin bundle enters the spine neck and binds to the PSD, which at the other end, is connected to the MTN. This is further detailed in Fig. 8.13, right, showing the actin bundle linked to the PSD.

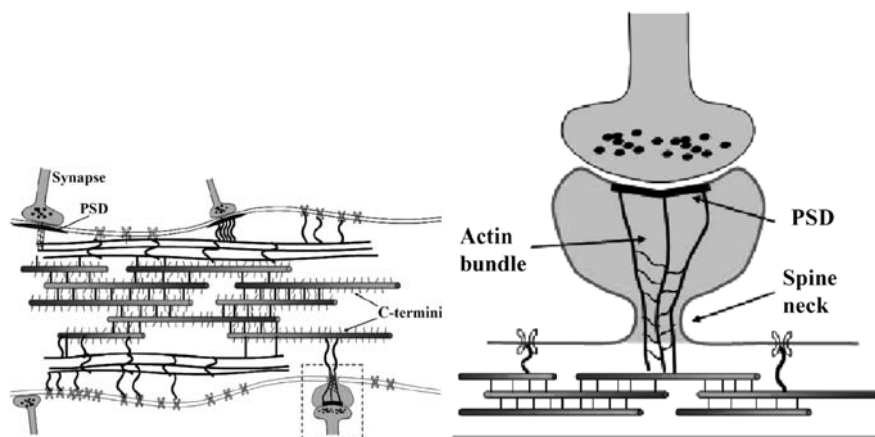


Fig. 8.13. *Left:* A scheme of the dendritic shaft. Depicted at the dendrite is a microtubule network, actin filaments, synapses and ion channels. Actin filaments enter the postsynaptic density (PSD). In the case of a spiny synapse (*lower right*) the actin bundle enters the spine neck. MTs are decorated by C-termini and interconnected by MAP2 (*bold lines*). The marked synapse is enlarged in the next figure. *Right:* Close-up view of the marked synapse in the *left figure*. Actin filaments enter the spine neck and connect to the PSD. The filaments are also connected to MTs (or MAP2s) at the dendritic shaft

We envision a mechanism in which a direct regulation of ion channels and thus synaptic strength by actin filaments and associated cytoskeletal structures controls and helps modify the electrical response of the neuron. In this picture, MTs arranged in networks of mixed polarity, receive signals in the form of electrical perturbations, from synapses via actin filaments connected to MTs by MAP2 [107], or via direct MT connections to postsynaptic density proteins by molecules such as CRIPT [94]. Due to the characteristics discussed earlier, the MTN may be viewed as a high-dimensional dynamic system where the main degrees of freedom are related to the (conformational) state of the C-termini. The input signals perturb the current state of the system that continues to evolve.

Hypothetical integration of the above ideas is outlined as follows:

1. Incoming (electrical) signals arrive at the postsynaptic density via synaptic transmission, which in turn elicit ion waves along the associated actin filaments at the synaptic spine (Fig. 8.14A).
2. These dendritic input signals propagate through actin filaments to the microtubules network (MTN) where they serve as input signals (Fig. 8.14B).
3. The MTN, operating as a large high-dimensional state machine, evolves these input states, e. g., by dynamically changing C-termini conformation (Fig. 8.14C).

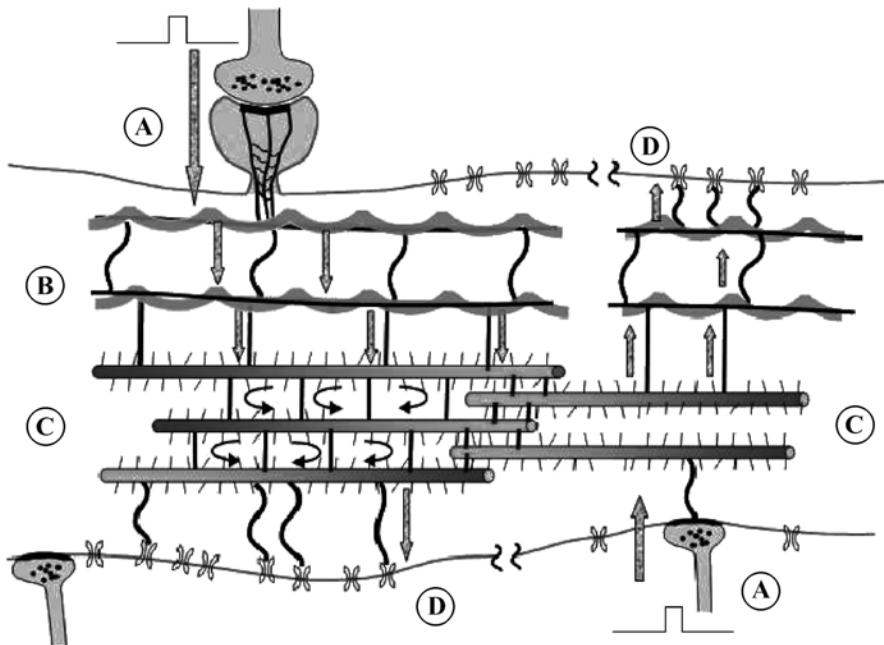


Fig. 8.14. Schematic picture of our model. **A** Incoming signal from the synapse is propagated via the actin bundle within the synaptic neck to the dendritic shaft, or directly to the MTN (see *lower-right* synapse). **B** The wave propagates along actin filaments (*thick wavy line*) and to a microtubule network (MTN). **C** The MTN operates as a high-dimensional dynamic system, which continuously processes the incoming perturbation. The curved arrows symbolize recurrent operations due to MAP2 coupling between adjacent MTs. **D** The current state of the “MTN state machine” is used as readout at certain areas, either directly to ion channels or mediated/propagated by the actin network, interacting with ion channels

4. The output from the MTN is the state of the system at some time “T” that is being “read” (sensed) in certain areas, again by actin filaments, and propagated to remote ion channels. These output functions are assumed to regulate the temporal gating state of voltage-sensitive channels (Fig. 8.14D).
5. This process subsequently regulates the membrane conductive properties as well as controlling the axon hillock behavior by changing the distribution and topology of open/closed channels. The overall function of the dendrite and neuron are thus regulated in this manner.

To understand how this integrative view may serve as a regulatory (adaptive) mechanism, let us elaborate on the main issues raised. The input level denoted by A in Fig. 8.14 is associated with external perturbations arriving through membrane processes. We mainly consider synaptic inputs arriving from other

neurons. These signals arrive at the cytoskeleton as electric perturbations, which either directly affect MTNs, and/or actin filaments (bundles), in spiny synapses. However, the actin cytoskeleton is responsible for the propagation of signals to the MTN (see level B). These propagated signals (level C) are in turn used as inputs to the MTN, which may be viewed as a dynamic system. The model further proposes that the MTN is capable of generating diverse phase-space trajectories in response to different input vectors. The requirements from such a system are in fact not too restrictive since the output state is not an attractor of the system. In other words, information processing at this level is not necessarily based on attractor dynamics but rather on real-time computations. This proposition relies on the observations regarding the ability of the MTN to propagate signals, on the one hand, and on the special (spatial) topological features of MTNs in dendrites, in which shorter MTs of mixed polarity are interconnected by MAP2s. The output from the MTN would be some function of the evolved state vector in certain areas accessed by actin filaments and/or directly linked to ion channels (see both possibilities at level D). These output signals may modify the temporal channel activity, either by directly arriving from the MTN to the channels, or mediated by actin filaments. Hence, the channel-based synaptic membrane conductance is regulated, in particular in the axon hillock region, which is, in most cases, responsible for the generation of action potentials.

The idea that a nonspecific high-dimensional dynamical system may serve as a reservoir of trajectories in the context of liquid state machines (LSM), has recently been suggested as a possible explanation for the existence of microcircuits in the brain [85]. The basic structure of an LSM is an excitable medium (hence “liquid”) and an output function that maps the current liquid state. The liquid must be sufficiently complex and dynamic to guarantee universal computational power. This may be sufficient to ensure that different input vectors will lead to separate trajectories. A network of spiking neurons, a recurrent neural network, and even real water have been used as a “liquid”, whereas the output (readout) function has been implemented by either simple perceptrons, threshold functions or even linear regression functions. Clearly, simpler readout functions restrict the ability of the whole system to capture complex nonlinear dependencies.

The attractiveness of the concept that the cytosol, with its cytoskeletal structures may behave as an LSM is obvious in the context of our model, because it provides a means for real-time computation without the need for stable attractors. Moreover, the output is relatively insensitive to small variations in the MTN (cytoskeletal networks!) or input vectors. Figure 8.15 depicts the functional scheme of the model described above and in Fig. 8.14. One should note that the MTN state evolves continuously, even without external inputs. Recent perturbations, however, have a long-term effect on the MTN trajectories, i.e. there is a memory effect inherent to this system (not to be mistaken with synaptic LTP, which possesses a much longer time scale).

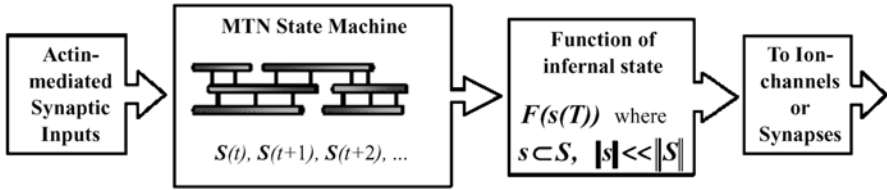


Fig. 8.15. Functional scheme of the model steps. Synaptic inputs are transferred to the MTN via actin filaments. These inputs perturb the MTN state machine that continuously evolves through time. The current state of the MTN in time “T” at certain areas is mapped by a function F and transmitted to its target, e.g., ion channels

The output from the MTN may be linear regression functions that converge at (or near) ion channels to regulate their temporal behavior. The issue of adapting the readout requires some sort of feedback mechanism that will, at least locally, enable the change in output function. In the context of neuronal function, with focus on processivity, synaptic strengthening, LTP, and memory enhancement, the output function may simply reflect an effect of the MTN on synaptic channel function, such that the desired state of the channel appears in a higher open probability. One possibility is a Hebbian-based response in which more frequent activity of certain subdomains of the MTN output states will give rise to higher (or lower, depending on the desired response) density of actin filaments connecting to corresponding channels.

8.4.1 MTN Control of Synaptic Plasticity, Modulation, and Integration

The above may reflect a most prominent feature of dendritic ion-channel regulation, with the potential boosting and enhancement of distal synaptic events, namely the interaction between channel activity and its regulation by cytoskeletal structures (Fig. 8.16). Voltage-gated Na^+ and Ca^{2+} channels play an important role in amplifying the magnitude of EPSPs towards the soma. Thus, channel regulation may also be important for dendritic interactions in the immediate vicinity of the synaptic input. Voltage-gated channels may alter the local input resistance and time constant, which in turn would influence both spatial and temporal summation of EPSPs and IPSPs, thus, conveying highly nonlinear interactions [122]; including signal amplification [46, 124]. The nonlinear properties of dendrites may act as amplification/modulation devices. In addition to providing an output, the action potentials elicited by synaptic input, whether initiated at the local synaptic sites or at the axon hillock and backpropagated into the dendrites, would essentially reset the dendritic membrane potential. This change in membrane potential would modify synaptic summation. Action potentials might also provide a feedback or an associative signal to other synapses active just before or just after the

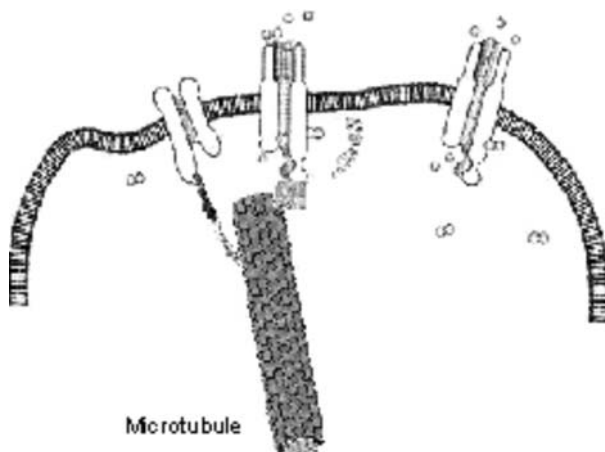


Fig. 8.16. MTs and actin filaments and interactions between the two (MTNs) elicit channel regulation in neurons. Conversely, synaptic channel function may also control cytoskeletal structures, thus structurally modifying the environment, and helping the dynamics of synaptic plasticity

action potential. Glutamate-activated NMDA receptors for example, could be rapidly opened by a backpropagating action potential that removes the voltage-dependent Mg^{2+} block of the channel [65, 125], thus producing Ca^{2+} influx through NMDA receptors. Feedback mechanisms between cytoskeletal structures (i. e. MTN) and synaptic signals may strengthen the adaptive behavior of the entire system.

Johnson and Byerly originally determined that agents that modify cytoskeletal organization alter Ca^{2+} -channel activity in *Lymnaea* neurons [47, 48]. Cytoskeletal disrupters such as colchicine and cytochalasin B were found to speed Ca^{2+} -channel decline in ATP, whereas the cytoskeletal stabilizers, taxol and phalloidin, were found to prolong Ca^{2+} -channel activity without ATP. In addition, cytoskeletal stabilizers reduced Ca^{2+} -dependent channel inactivation. Thus, both channel metabolic dependence and Ca^{2+} -dependent inactivation in neurons might be controlled by cytoskeletal interactions. Further, Ca^{2+} -channel inactivation in acutely isolated rat hippocampal pyramidal neurons [48] is modified by drugs that stabilize (taxol and phalloidin) and destabilize (colchicine and cytochalasin B) the actin and microtubular cytoskeletons, respectively. Voltage-gated K^{+} channels from retinal bipolar neurons of the tiger salamander [74] were also found to be controlled by acute disruption of endogenous actin filaments with cytochalasin D, which was largely prevented by intracellular perfusion with the actin filament-stabilizer agent, phalloidin. Thus, acute changes in actin-based cytoskeleton dynamics also regulate voltage-gated ion-channel activity in bipolar neurons. This may be indicative of a more general and quite appealing mechanism by which cytoskeletal structures control feedback mechanisms in neuronal channels. Cy-

toskeletal remodeling may help sort, or otherwise deliver ion channel subunits to several neuronal locations. Cytoskeletal control of neuronal channels may also target channel localization and clustering. The discrete location of Na^+ channels to subdomains of the nerve, suggested that the Nav1.2 voltage-gated Na^+ channel contains axonal sorting and/or clustering signals [34]. Evidence indicated that addition of the cytoplasmic carboxy-terminal region of Nav1.2 restricted the distribution of a dendritic–axonal reporter protein to axons of hippocampal neurons. Interestingly, this motif was also sufficient to redirect somatodendritic K^+ channels to the axonal initial segment, a process involving association with ankyrin G.

Conversely, electrical synaptic activity may have the opposite effect in the control of subjacent cytoskeletal structures [30] (Fig. 8.16). Afferent back-propagating action potentials provide a feedback to dendritic synapses after a neuronal output has occurred. Ca^{2+} channels are implicated in postsynaptically induced synaptic plasticity such as short-term potentiation (STP) [75], long-term potentiation (LTP) (both NMDA-receptor dependent and NMDA-receptor independent), and long-term depression (LTD) [8, 14, 50]. This, in turn, is highly dependent on the control of the cytoskeletal structures involved in their vicinity [78, 79]. LTP induction is associated with a long-lasting increase in actin filament content within dendritic spines. This phenomenon, however, is dependent on NMDA receptor channel activity [32]. Similarly, long-term facilitation, and the response to cAMP-induced synaptic enhancement are both dependent on I_h -induced presynaptic current regulation of the actin cytoskeleton [5]. Thus, electrical synaptic activity is an essential component of structural stability and dynamic function of cytoskeletal structures, which in turn feedback the membrane components of the dendritic spine, to name a central player in synaptic plasticity. It is obvious that other neuronal regions are similarly affected, and may be dynamically regulated by such dual interactions.

8.5 Final Statement

In this report we provide extensive evidence for new roles of the actin and microtubule-based cytoskeletons in the control of dendritic function and morphological features. The most novel information relies on our findings that both structural elements of their respective cytoskeletons; namely actin filaments, and microtubular structures (MT-MAPs), support ionic waves, which may provide a new dimension to the control of neural function.

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